

UWB Indoor Positioning System

Complete Working Principle · Project **uwbtt2** (SK TechChange) · Jun 21, 2026

System Overview

A real-time indoor positioning system using **UWB (Ultra-Wideband)** radio via the **Qorvo DW3000** chip on **ESP32-S3** boards. The system is organized into three tiers:

Hardware (Tag + Anchors) → Cloud (Firebase) → Web (Browser)

1. Hardware Tier (Edge)

Tag — the mobile unit (initiator)

- ESP32-S3 + DW3000 carried by a person or attached to the object to be tracked.
- Ranges to **Anchor 1** → **2** → **3** one at a time each loop using **SS-TWR (Single-Sided Two-Way Ranging)**.
- SS-TWR principle: the Tag sends a signal → the Anchor replies → the Tag measures the **round-trip time** and computes distance from the speed of light.
- Produces distances **d1, d2, d3** (metres) and uploads them to Firebase at **/live** over WiFi (~3 times/second).

Anchors — three fixed units (responder)

- The **same firmware** is flashed to all 3 boards; only **ANCHOR_ID** (= 1, 2, 3) differs per board.
- Main role: continuously respond to the Tag's ranging requests.
- Special role: **web-triggered calibration** — anchors range each other (**d12, d13, d23**) so the system learns the anchor layout without manual measurement.

2. Calibration Principle (the standout feature)

Instead of measuring anchor distances by hand and typing them in, the system uses **web-triggered calibration**:

- Press **"Calibrate anchors"** in the web app → the app writes a fresh "token" to **/control/calibrate**.
- Every anchor **polls that token every 3 seconds** (**CALIB_POLL_MS**).
- On seeing a new token → each anchor ranges its peers **once** and **PATCH**es **/calib**.
- Who initiates: the lower-ID anchor of each pair → A1 measures **d12, d13** · A2 measures **d23** · A3 only responds.
- Since all anchors get the token at once, each **staggers its start** by **(ID-1) × 1.2 s** so the initiator can range a peer while that peer is still listening.

Important: At boot, an anchor adopts whatever token already exists *without* calibrating → it never calibrates on power-up. Values persist in Firebase, so you only press the button again if the anchors physically move.

3. Cloud Tier (Firebase)

The Firebase Realtime Database acts as the **central message bus** — every tier communicates by reading/writing this database; nothing talks peer-to-peer over IP.

Path	Written by	Data
/live	Tag	{d1, d2, d3, seq, ts} — tag→anchor distances
/calib	Anchors	{d12, d13, d23, ts} — anchor-to-anchor distances
/control/calibrate	Control site	token (number) — calibration request
/display	Control site	config for the viewer (zones, geometry, options)

In addition, **Firebase Hosting** serves the static `web/` folder over CDN at `https://uwb-positioning-a2892.web.app`.

Dropout handling: If ranging is briefly lost, the system **holds the last good distance** (up to ~9 s, `RANGE_HOLD_MAX`) so the position stays steady; only a prolonged loss is reported as `-1` ("no data").

4. Web Tier (Frontend) — where the real positioning math runs

The web app is plain HTML/JS with **no build step**, split into two sites:

Control site (`index.html` / `app.js`)

- All setup: enter the Firebase host, trigger calibration, set geometry, view telemetry.
- Includes a **zone editor** to draw trigger areas on the map, then publishes everything to `/display`.

Viewer site (`viewer.html` / `viewer.js`)

- A **read-only** site for a wall display or tablet (titled "SK TechChange").
- Reads `/live`, `/calib`, `/display` to render the live map and pop-up notifications.

The core math — Trilateration (runs in the browser)

The database only stores *distances*, never *coordinates* — the browser computes positions itself:

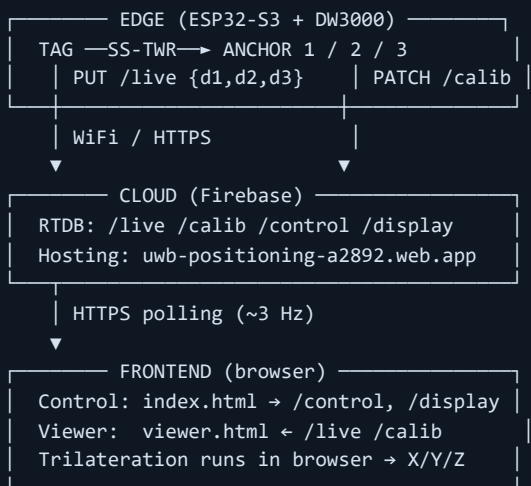
- From `/calib`, derive the anchor layout: place $A1=(0,0)$, $A2=(d12,0)$, then solve $A3$ from $d13$ / $d23$.
- From `/live` (d1, d2, d3), trilaterate the tag's **X / Y / Z** position.
- Display: X/Y/Z values, a fit residual (lower = better), Top view (X-Y) and Side view (X-Z), plus EMA smoothing to tame jitter.

Zone triggers — notification areas

- Draw box areas on the map and set name/message/colour/icon/beep/image/audio-guide/link.
- When the tag enters an area → a pop-up appears, an optional beep sounds, and the event is logged.
- The default set in `zones.config.js` is the **SK TechChange** tour — 6 attractions with images + MP3s.

About Z (height) ambiguity: Three anchors define a **flat plane**, so the computed Z could be the same distance above or below it. Use the "**Tag is below anchor plane**" toggle to pick the side. For robust 3-D, add a **4th anchor mounted out of the plane**.

Full System Flow



1. Tag ranges 3 anchors → writes /live
2. (On calibrate) Anchors range each other → write /calib
3. Control site writes config → /display
4. Browser reads everything → solves geometry + trilateration → renders map + zone alerts

Everything is connected only through Firebase, cleanly decoupling the hardware, cloud, and web tiers.